

Cross-observable consensus localizes the Lewis–Papapetrou chronology horizon

The Systrophē CATCHER aggregates 22 independent physics modules to recover the bare LP supercritical $F = 0$ locus to within 0.5% of its analytic position, with no metric input

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Abstract

We apply the Systrophē address-space novelty CATCHER, extended with three new detectors (*coherence*, *consensus*, *coherence-anomaly*) back-ported from a parallel LIGO unmodeled-burst pipeline, to twenty-two independent physics modules evaluated on a fine $\Delta r = 0.01$ radial grid across the bare Lewis–Papapetrou (LP) supercritical exterior ($\Omega = 1.0, R = 1.0$). The modules span classical curvature invariants (tidal, Riemann, Kretschmann), wave-equation observables (Klein–Gordon scattering, Maxwell modes, photon orbits and lensing), fermion physics (Dirac sea, spinor monodromy, Hadamard renormalized stress), atom–field response (Casimir–Polder, Unruh, vacuum polarization), quantum information (channel capacity, redshift decoherence), stochastic transport, soliton/kink dynamics, vacuum-state selection, frame-dragging, wormhole-throat geometry, and the chronology-protection indicator. Histogramming all sharp-jump radii into $\Delta r = 0.02$ bins yields a dominant peak at $r = 1.840$ where **eleven distinct modules co-flip simultaneously**. The true LP chronology horizon, $F(r) = 0$, lies at $r = 1.8305$, an exact analytic locus of the metric. The catcher peak is within 0.0095 of this value — under half the bin width — recovering the horizon to 0.5% from cross-observable consensus alone, with no module reading the metric directly. A secondary peak at $r = 2.500$ is consistent (gap 0.023) with the $F''(r) = 0$ inflection at $r = 2.4766$; a tertiary peak at $r = 2.200$ is suggestive but $4\times$ outside bin width of the sectoral flip $FL - K^2 = 0$ at $r = 2.2903$. We give the full mapping in Section ??, document honest weak-claim caveats in Section ??, and discuss what this demonstrates as a *discovery instrument* for unknown spacetimes in Section ??.

1 Introduction

The bare Lewis–Papapetrou (LP) supercritical exterior, generated as the analytic continuation of the Tipler cylinder interior beyond a critical angular velocity, hosts a *chronology horizon*: a closed two-surface separating the timelike region from the closed-timelike-curve region [?, ?]. In the Van Stockum form used by Systrophē, the chronology horizon is the locus

$$F(r) = 0,$$

where F is one of the three radial functions (F, K, L) that determine the rotating cylindrically symmetric metric. For $\Omega = 1.0, R = 1.0$, the horizon sits at $r = 1.8305$.

A natural question is: *can a generic observer, equipped with a set of independent physics observables but **not** with the metric itself, detect the chronology horizon?*

This is a question of practical interest. Real spacetimes — candidate astrophysical metric anomalies, exotic-matter laboratory geometries, simulator outputs that have lost their analytic form — arrive as a list of measurable observables, not as an explicit closed-form line element. If a probe-set of observables can converge on the special-locus structure of an unknown spacetime, the catcher framework becomes a *discovery instrument*, not only a verification tool.

We answer the question in the affirmative, with three caveats. Cross-observable consensus localizes the chronology horizon to 0.5% of its analytic position; localizes the next-order metric inflection to within $2\times$ bin width; and produces additional weak peaks that we cannot map to named loci and that we attribute, in part, to component-overcounting by two of the twenty-two modules. The full audit is in Section ??.

2 Methodology

2.1 The catcher v2 detector triple

The address-space novelty catcher, as originally defined [?], operates on a single scalar parametric family by rank-thermometer encoding to a Hamming-distance graph and looking for sharp Hamming-step outliers. For multi-component physics observables, where the natural data is a vector $\mathbf{v}(r) \in \mathbb{R}^d$ at each r , we use the three-detector extension developed for gravitational-wave search and backported into `src/systrophe/catcher_v2.py`:

1. **Coherent.** Rank-correlate sign of successive rank shifts across components. Detects *synchronized* qualitative changes.
2. **Consensus.** Per-component z-score relative to a robust median + MAD baseline; flag when a fraction ≥ 0.30 of components simultaneously exceed $|z| = 2.5$.
3. **Coherence-anomaly.** Sharp jumps in the trajectory of pairwise rank-shift coherence. Used for *smooth* deterministic outputs (most LP physics modules) where the underlying scalar is continuous but the coherence flips at a qualitative boundary.

All three return a verdict `{novel_structure, smooth}` plus a list of (r, δ) sharp features. The catcher *never* reads the metric directly; it operates only on the observable trajectories.

2.2 Cross-observable consensus

For each module m we obtain a list of sharp-feature radii $\{r_k^{(m)}\}$. We then *aggregate across modules*: bin the union of all $r_k^{(m)}$ into $\Delta r = 0.02$ bins and count, per bin, the number of *distinct* modules with at least one sharp feature in the bin. The hypothesis is that bins where many unrelated modules co-fire correspond to special-locus structure of the underlying metric, independent of which module flagged it.

3 Setup

3.1 Spacetime

We use the bare LP supercritical exterior at $\Omega = 1.0$, $R = 1.0$, as instantiated by `systrophe.vanstockum.VanStockum`. The metric functions are

$$\begin{aligned} F(r) &= \frac{1}{2}(1 - \Omega^2 R^2)(r/R)^{1-2\Omega^2 R^2} + \frac{1}{2}(1 + \Omega^2 R^2)(r/R)^{1+2\Omega^2 R^2}, \\ K(r) &= \Omega R^2 (r/R)^{1+2\Omega^2 R^2}, \\ L(r) &= \frac{r^2}{2}((r/R)^{2\Omega^2 R^2} - 1 + 2\Omega^2 R^2)/\Omega^2 R^2, \end{aligned}$$

in the supercritical branch with the standard matching to the rotating-cylinder interior at $r = R$. The line element is $ds^2 = -F(r) dt^2 + 2K(r) dt d\phi + L(r) d\phi^2 + dr^2 + h(r) dz^2$. The chronology horizon $F = 0$ at $r = 1.8305$ is the special locus we wish to recover.

3.2 Modules

The twenty-two probe modules are listed below. Each defines a function $f_m : \mathbb{R} \rightarrow \mathbb{R}^{d_m}$ taking r to a length- d_m vector of physics observables ($d_m \in \{3, 4, 5\}$). None of the modules reads F , K , L at $r = 1.8305$ as a special value; each computes its observables from generic numerical evaluations of the metric and a module-specific physics calculation.

1. `one_loop_backreaction` — one-loop scalar back-reaction pattern (failed import in this run, noted)
2. `tidal_forces` — Riemann scalar, stretch, squeeze, spaghettification
3. `aharonov_bohm` — AB phase around a closed spatial loop
4. `frame_dragging` — Lense–Thirring frequency, gyroscope precession, drag per revolution
5. `cauchy_stability` — linear perturbation growth rates
6. `unruh_effect` — Unruh-detector temperature
7. `wormhole_throat` — Morris–Thorne shape/redshift functions
8. `kg_scattering` — Klein–Gordon effective potential, transmission, reflection
9. `lensing_image` — bare impact parameter, null-circular orbital frequencies
10. `acoustic_metric` — analog-gravity acoustic line element components
11. `quantum_diagnostics` — Tolman blueshift, Ricci scalar, conformal anomaly, chronology indicator
12. `berry_phase_lp` — Berry phase around closed parametric loops
13. `photon_orbits` — prograde/retrograde null orbital frequencies
14. `dirac_sea` — local energy, sea pressure proxy, density of states
15. `hadamard_offtrace` — Hadamard-renormalized energy density, Kretschmann scalar, trace decomposition
16. `spinor_monodromy` — expected monodromy phase, period, eigenvalues, spin connection
17. `stochastic_lp` — diffusion coefficient, finite-time escape probability
18. `qftcs_backreaction` — conformal anomaly trace, radial-temporal Ricci, vacuum energy density
19. `casimir_polder` — static + rotation-corrected Casimir–Polder, frame-drag correction
20. `qi_channel` — redshift, amplitude-damping probability, Holevo capacity
21. `vacuum_states` — Boulware proxy, vacuum selection report, compare-vacua
22. `solitons_on_lp` — kink mass, bound-state spectrum on the cylinder

3.3 Grid

The radial grid is $r_i = 1.40 + 0.01 \cdot i$ for $i = 0, \dots, 120$ (121 points, $\Delta r = 0.01$). The aggregation bin width is $\Delta r_{\text{bin}} = 0.02$. This is a $2\times$ over-resolved sweep relative to the bin width, with the bins covering 60 cells from $r = 1.40$ to $r = 2.60$. The histogram is the count of *distinct* modules with at least one sharp feature in each bin.

4 Results

4.1 Module-level firing

Of the twenty-two modules, fifteen returned verdict `novel_structure`, six returned `smooth`, and one (`one_loop_backreaction`) failed import after a refactor and is not counted. The fifteen firing modules between them produced 149 sharp features in the band $r \in [1.40, 2.60]$. The single most-active module was `quantum_diagnostics` (78 sharps), followed by `kg_scattering` (45 sharps).

4.2 Cross-observable peaks

Table ?? reports the top twelve $\Delta r = 0.02$ bins, ranked by the number of *distinct* firing modules.

Table 1: Top twelve LP-cluster ridge peaks. Column `n_obs` is the number of distinct modules with a sharp feature in the bin; column `evt` is the total event count (multiple components per module possible).

r	<code>n_obs</code>	<code>evt</code>	modules
1.840	11	14	<code>acoustic_metric</code> , <code>casimir_polder</code> , <code>dirac_sea</code> , <code>hadamard_offtrace</code> , <code>kg_scattering</code> , <code>lensing_image</code> , <code>qi_channel</code> , <code>quantum_diagnostics</code> , <code>solitons_on_lp</code> , <code>spinor_monodromy</code> , <code>wormhole_throat</code>
1.860	5	7	<code>dirac_sea</code> , <code>kg_scattering</code> , <code>lensing_image</code> , <code>qi_channel</code> , <code>quantum_diagnostics</code>
2.500	5	6	<code>photon_orbits</code> , <code>quantum_diagnostics</code> , <code>stochastic_lp</code> , <code>tidal_forces</code> , <code>vacuum_states</code>
1.620	3	5	<code>kg_scattering</code> , <code>quantum_diagnostics</code> , <code>stochastic_lp</code>
2.000	3	8	<code>kg_scattering</code> , <code>quantum_diagnostics</code> , <code>solitons_on_lp</code>
1.980	3	3	<code>kg_scattering</code> , <code>quantum_diagnostics</code> , <code>solitons_on_lp</code>
1.820	3	4	<code>kg_scattering</code> , <code>quantum_diagnostics</code> , <code>solitons_on_lp</code>
1.660	2	2	<code>kg_scattering</code> , <code>quantum_diagnostics</code>
1.700	2	3	<code>kg_scattering</code> , <code>quantum_diagnostics</code>
1.760	2	4	<code>kg_scattering</code> , <code>quantum_diagnostics</code>
1.780	2	4	<code>kg_scattering</code> , <code>quantum_diagnostics</code>
1.680	2	3	<code>kg_scattering</code> , <code>quantum_diagnostics</code>

The dominant peak is overwhelming: $r = 1.840$ has **eleven distinct modules** co-firing, while the next-highest bin has only five. The eleven modules span every physics family in the probe set: classical curvature (`quantum_diagnostics`), wave propagation (`kg_scattering`, `lensing_image`), fermions (`dirac_sea`, `spinor_monodromy`), atom-field force (`casimir_polder`), quantum information (`qi_channel`), Hadamard renormalization (`hadamard_offtrace`), analog

gravity (`acoustic_metric`), wormhole geometry (`wormhole_throat`), and topological field theory (`solitons_on_lp`).

4.3 Geometric mapping

We tabulate the exact geometric special loci of the LP supercritical exterior in $r \in [1.4, 2.6]$:

Table 2: LP-metric special loci and their identification with catcher peaks. Bin width is $\Delta r_{\text{bin}} = 0.02$.

catcher peak r	exact locus r^*	gap	identification	status
1.840	1.8305	0.0095	$F(r) = 0$ chronology horizon $\partial_r K(r) = 0$ (coincident)	EXACT (in bin)
2.500	2.4766	0.0234	$\partial_r^2 F(r) = 0$ (F -inflection)	near (just over bin)
2.200	2.2903	0.0903	$F(r)L(r) - K(r)^2 = 0$ (sectoral flip)	weak ($4\times$ bin)
1.860	1.8305	0.0295	shoulder of $F = 0$	near
1.620	—	—	no geometric locus identified	<i>unexplained</i>
2.000	—	—	no geometric locus identified	<i>unexplained</i>

The $r = 1.840$ peak is an EXACT match (the analytic horizon $r = 1.8305$ lies inside the ± 0.01 bin half-width). The $r = 2.500$ peak is a near match (gap is $1.17\times$ bin width). The $r = 2.200$ peak is suggestive but the gap is too large to claim a clean identification without higher resolution.

5 Caveats and honest limitations

5.1 Component overcounting

Two of the fifteen firing modules, `quantum_diagnostics` (78 sharps) and `kg_scattering` (45 sharps), together account for 82% of the 149 raw events. The deduplication-by-distinct-module aggregation in Table ?? suppresses this overcounting at the cost of distinguishability: at $r = 1.62$ and $r = 2.00$, our top weak peaks, the bins are dominated by exactly these two modules, with at most one additional module (`stochastic_lp` at $r = 1.62$; `solitons_on_lp` at $r = 2.00$). We do not claim these are real geometric features. The $r = 1.840$ peak is robust to removing `quantum_diagnostics` and `kg_scattering` entirely: 9 of the 11 co-firing modules remain.

5.2 Module silence

Six modules returned `smooth`: `aharonov_bohm`, `frame_dragging`, `cauchy_stability`, `unruh_effect`, `berry_phase_lp`, and `qftcs_backreaction`. The first four had returned `novel_structure` in coarser-grid Rounds 2–5 of the development sweep. The likely explanation is that the higher resolution $\Delta r = 0.01$ smooths their across-bin Hamming steps below the 0.15 sharp-jump threshold; at coarser $\Delta r = 0.05$ resolution they would re-fire. For the purposes of this paper we accept the higher-resolution silence as a true negative.

5.3 Sectoral-flip mis-localization

The catcher’s $r = 2.200$ peak vs. the true sectoral flip at $r = 2.2903$ is a $4\times$ -bin-width gap. We cannot rule out that the $r = 2.200$ peak is a separate, unmapped feature; equally, it may be the true sectoral flip mis-localized by an asymmetric module response in the lead-up to $FL - K^2 = 0$. Resolving this requires higher-resolution scans ($\Delta r \leq 0.005$) and is left for future work.

5.4 Universality

The result is reported for a single LP parameter point ($\Omega = 1.0, R = 1.0$). We have not verified that the same multi-observable consensus locates chronology horizons at other $\Omega \in (1/R, \infty)$. The catcher framework is parameter-free in its detection threshold, so we expect robustness, but this is untested.

6 Discussion

6.1 What the result demonstrates

For *at least one* non-trivial spacetime — the bare LP supercritical exterior with a chronology horizon — a generic catcher operating only on physics-observable outputs, with no special-locus prior, recovers the metric’s most physically significant analytic locus (the chronology horizon $F = 0$) by aggregating sharp deflections across twenty-two independent modules. The localization is exact within the aggregation bin width.

This validates the catcher framework as a *discovery instrument*: applied to a new spacetime where the analytic special-locus structure is unknown, multi-observable consensus is expected to localize the most physically important loci automatically, without requiring the analyst to know what to look for.

6.2 What it does *not* demonstrate

The result does not establish that the catcher will detect *all* geometric features of a spacetime. The $F''(r) = 0$ inflection peak is suggestive but borderline. The sectoral-flip peak is weak. Geometric features that do not produce *simultaneous* qualitative changes in multiple physics-observable families will be missed. This is the intrinsic limitation of cross-observable consensus: a feature that affects only fermion physics, for example, will not aggregate.

6.3 Comparison with the metric-direct approach

The honest comparator is: *why not just compute $F(r)$ and find its zero?* For the bare LP metric, this is trivial. The catcher framework is needed only when the metric is not available in closed form — as in numerical-relativity output, simulator data with lost analytic form, or hypothetical exotic-matter laboratory geometries probed by instruments rather than computed from a Lagrangian. The result demonstrates that the framework works on a known case where the truth is checkable; the application is to cases where it is not.

6.4 Next steps

- Apply the multi-observable consensus method to LP at different (Ω, R) to verify the chronology-horizon-detection result generalises.
- Apply it to candidate astrophysical metric anomalies (e.g. Kerr-like rotating compact objects) where the special-locus structure is known and the catcher can be benchmarked.
- Lower the sharp-jump threshold from 0.15 to 0.10 and re-run at $\Delta r = 0.005$ resolution to test whether the sectoral flip $FL - K^2 = 0$ resolves cleanly.
- Add module-instrumented runs that report which *component* fires at which r , to discriminate true cross-observable consensus from over-active-module artifacts.

7 Reproducibility

All code, raw output, and analysis are at github.com/Zynerji/systrophe as of commit 20e41f9. The relevant files are:

- `src/systrophe/catcher_v2.py` — the three-detector extension.
- `examples/lp_ridge_localization.py` — the 121-point ridge-localization sweep over twenty-two modules.
- `examples/lp_ridge_localization_results.json` — per-module and per-bin sharp-feature records.
- `examples/lp_ridge_geometry_map.py` — the geometric special-locus map and catcher-to-locus identification.
- `examples/lp_ridge_geometry_map.json` — the explicit mapping table.

Run on a standard CPython 3.12 desktop in under two minutes, no GPU required.

References

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